

PYTHON SYLLABUS		
OBJECTIVE: The objective of the training session is to impart knowledge on PYTHON programming which is used on a server to create web applications. The training session also includes how to work alongside software to create workflows and connect database systems and perform complex mathematics for rapid prototyping.		
Duration:		
S:NO	CONTENT	HOURS (L/P)
1	Introduction to Python	
	Course Overview	
	History and Features of Python	
	Setting Up the Development Environment (Installing Python, IDEs)	
	Writing and Running Your First Python Program	
	Python Basics: Syntax, Indentation, and Comments	
2	Variables and Data Types	
	Variables and Assignments	
	Basic Data Types (int, float, str, bool)	
	Type Conversion	
3	Operators and Expressions	
	Arithmetic Operators	
	Comparison Operators	
	Logical Operators	
	Assignment Operators	
4	Control Structures	
	Conditional Statements (if, elif, else)	
	Loops (for, while)	
5	Functions	
	Defining and Calling Functions	
	Function Parameters and Return Values	
	Variable Scope (local and global variables)	
	Lambda Functions	
6	Data Structures - Lists and Tuples	
	Lists: Creation, Indexing, Slicing, Methods	
	List Comprehensions	
	Tuples: Creation, Indexing, Slicing	
7	Data Structures - Sets and Dictionaries	
	Sets: Creation, Methods, and Operations	
	Dictionaries: Creation, Accessing Values, Methods	
	Dictionary Comprehensions	
	Use Cases for Sets and Dictionaries	
8	Strings and String Operations	
	String Creation and Indexing	
	String Methods and Formatting	
	String Slicing	
	Regular Expressions (Basics)	

9	File Handling	
	Reading from and Writing to Files	
	File Methods and Context Managers (with statement)	
	Working with Different File Types (text, CSV, JSON)	
10	Exception Handling	
	Introduction to Exceptions	
	Try, Except, Finally Blocks	
	Raising Exceptions	
11	Modules and Packages	
	Importing Modules and Using Standard Libraries	
	Creating and Using Custom Modules	
	Introduction to Packages	
	Using pip for Package Management	
12	Object-Oriented Programming (OOP)	
	Classes and Objects	
	Attributes and Methods	
	Inheritance and Polymorphism	
13	Working with Libraries	
	Overview of Popular Python Libraries (NumPy, Pandas, Matplotlib, etc.)	
	Basics of NumPy for Numerical Computations	
	Basics of Pandas for Data Manipulation	
	Basics of Matplotlib for Data Visualization	